

Debugging Aliasing After Unfiltered Decimation

A 24 kHz signal is decimated by 4 with no anti-aliasing filter, and the output spectrum shows aliasing.

24 kHz

fs (in)

÷ 4

Decimation

6 kHz

fs (out)

3 kHz

New Nyquist

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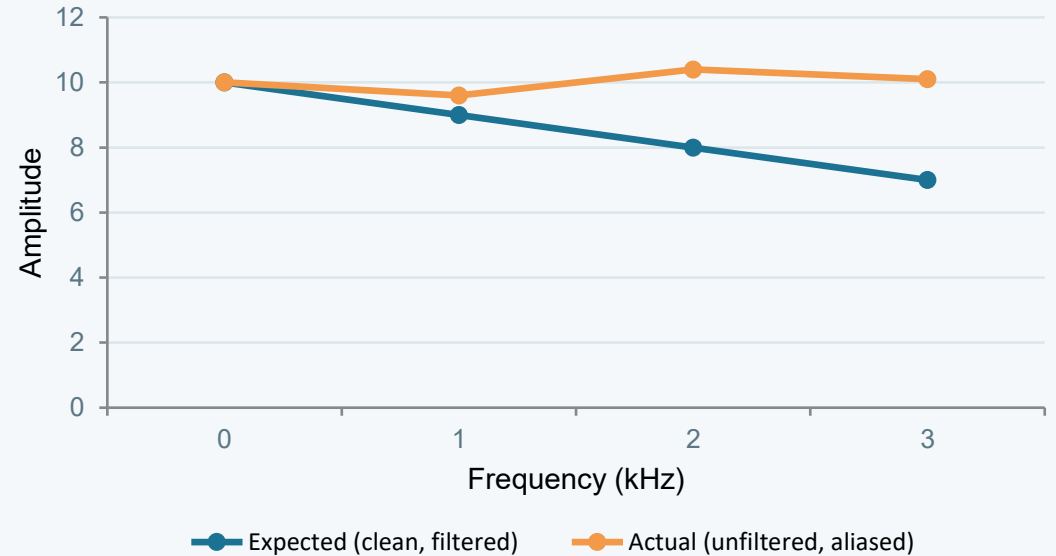
WHAT WENT WRONG

Understanding the Problem

- $f_s = 24 \text{ kHz} \rightarrow$ original Nyquist = 12 kHz.
- Decimate by 4 (keep 1 of every 4 samples) $\rightarrow$ new $f_s = 6 \text{ kHz} \rightarrow$ new Nyquist = 3 kHz.
- No low-pass filter is applied before the sample drop.
- Any content between 3 kHz and 12 kHz folds back (mirrors) into the 0–3 kHz band.
- Output spectrum shows extra, unexplained energy — the classic aliasing signature.

Why it matters: aliased energy is indistinguishable from real signal — it silently corrupts every downstream measurement.

Output Spectrum (0–3 kHz): Expected vs Actual



Symptom to look for: the actual curve bulges above the expected curve — energy that shouldn't exist at low frequency.

Confirm the Cause

1

Recompute the new Nyquist frequency

$fs_{\text{new}} = fs_{\text{in}} / M = 24 \text{ kHz} / 4 = 6 \text{ kHz} \rightarrow \text{Nyquist} = fs_{\text{new}} / 2 = 3 \text{ kHz}.$

2

Check the pipeline for a filter stage

Search the code/block diagram for a low-pass (anti-aliasing) filter placed before the downsampler. If it's missing, that's the prime suspect.

3

Plot the input spectrum before decimation

Run an FFT on the original 24 kHz signal and mark everything above 3 kHz — this is the content at risk of aliasing.

Checkpoint: if signal content exists above the new Nyquist and no filter is found, aliasing is expected — proceed to confirm it in the spectrum.

Isolate the Aliasing

4

Compare input vs. output spectra side by side

Overlay the pre-decimation FFT (0–12 kHz) with the post-decimation FFT (0–3 kHz).

5

Look for the folding / mirror pattern

Energy from $(3 + \Delta f)$ kHz in the input reappears at $(3 - \Delta f)$ kHz in the output — the tell-tale aliasing signature.

6

Rule out other causes

Confirm sample count, indexing, and filter coefficients elsewhere are correct, so the fault is isolated to the missing anti-aliasing filter.

Root cause confirmed: decimation by 4 with no anti-aliasing low-pass filter before the sample drop.

THE FIX

Add the Filter, Then Verify

- Insert a low-pass (anti-aliasing) filter before the decimator.
- Set the cutoff at or below the new Nyquist: ≤ 3 kHz.
- Use an FIR (linear-phase) or CIC filter matched to a decimate-by-4 stage.

Re-run decimation → re-plot the output FFT → confirm the fix.

Corrected Signal Chain



Verification Checklist

- ✓ Anti-aliasing filter present before downsampling
- ✓ Cutoff frequency $\leq$ new Nyquist (3 kHz)
- ✓ Filter order sufficient for required stop-band attenuation
- ✓ Output spectrum has ~ 0 energy above 3 kHz
- ✓ No mirrored / folded components remain